

DEVITLTD. Event Management System

MERN Stack Project Documentation

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- Project Overview

DEVIT EVENT is a full-stack Event Management and Reservation Platform developed using the MERN Stack. The platform connects organizers and attendees through a responsive web application. Organizers can create events, manage ticket categories, monitor bookings, and view analytics. Attendees can browse events, filter by category or city, book tickets, receive QR-based digital tickets by email, and access booking history.

- Main Features

User registration and login with JWT authentication

Dynamic city and category filters

Event detail pages with schedules and ticket availability

Normal, VIP and VVIP ticket booking

QR ticket generation and instant email delivery

Organizer analytics dashboard

Event creation and editing

Admin moderation and booking reset utilities

- Technology Stack

Frontend: React.js (Vite)

State Management: Redux

Toolkit Backend: Node.js + Express.js

Database: MongoDB Atlas

Authentication: JWT + BCryptJS

Email: Nodemailer

QR Code: QRServer

API Styling: CSS

Version Control: Git or GitHub

- Project Architecture

React Frontend □ REST APIs (Node + Express) □ MongoDB Atlas

Additional service: Nodemailer sends booking confirmation emails with QR tickets.

- Database Schema

Users Collection: _id, name, email, password, role, createdAt

Events Collection: title, description, prices, category, location, date, time, availableSeats

Bookings Collection: user, event, quantity, totalPrice, ticketType, createdAt

- API Endpoints

Authentication:

POST /api/v1/auth/register POST /api/v1/auth/login

Events:

GET /api/v1/events GET /api/v1/events/:id POST /api/v1/events PUT /api/v1/events/:id

DELETE /api/v1/events/:id

Bookings:

POST /api/v1/bookings

GET /api/v1/bookings/my-bookings

GET /api/v1/bookings/organizer-bookings DELETE /api/v1/bookings/reset

- Authentication Flow

User submits credentials □ server verifies □ JWT token generated □ stored in sessionStorage

❑ Axios sends Authorization header ❑ middleware validates ❑ access granted.

• Folder Structure

```
event-management/  
  
App/  
├ public/  
├ src/  
│ ├ components/  
│ ├ pages/  
│ ├ routes/  
│ ├ store/  
│ └ utils/  
  
backend/  
├ config/  
├ controllers/  
├ middleware/  
├ models/  
├ routes/  
└ services/  
  
package.json  
README.md
```

• Final Output

DEVIT EVENT was successfully developed as a complete MERN stack application demonstrating frontend development, REST APIs, MongoDB integration, authentication, QR generation, and deployment.

- GitHub Repository : [DEVITLTD](#)
- [_Live Demo](#) : [DEVITLTD](#)